TatvaAI: feature list in simple words

Follows the [feature map](docs/FEATURE_MAP.md).

What the project does

Every year, colleges ask AICTE to renew their approval. They upload documents like fire safety and land certificates. Officers check them by hand, which is slow, and fake or expired papers can slip through.

TatvaAI reads those documents, checks them, and warns the officer about problems. It shows proof for every warning. It gives each college a risk colour: GREEN (no risk), AMBER (medium risk) or RED (high risk). It also makes a list of which colleges inspectors should visit first.

The software never approves or rejects anyone. A human officer decides, and must write a reason.

Status words: **Done** = works for real and is covered by tests.

The three main features

Feature 1: Handbook rule checker

Feature	What it means	Status
Teachers, rooms, computers, bandwidth	Checked against the AICTE Handbook numbers	Done
Built-up area	Declared building area compared with the buildings on the map	Done
Certificate dates	Expired certificates, or ones that run out before the academic year starts	Done
Clause and page on every warning	Each warning says which Handbook rule it comes from	Done
Rules in files	The norms are editable files, not hidden in code	Done

Feature 2: Document & consistency check

Feature	What it means	Status
Genuine document check	Sees if a document was changed after it was signed, and shows what changed; QR codes compared with the printed text	Done
Source check	Asks the issuing source (fake e-stamp, DigiLocker, CA, fire, building, AICTE systems, all labelled MOCK) if the document is real	Done
Six answers	Verified, mismatch, expired, pending, source down, needs manual check	Done
Facts agree	The same fact is compared across documents, years, colleges and the college website	Done
Ghost teachers	A teacher listed full-time at two colleges is caught	Done
Copied photos	Reused or far-away photos are caught	Done

Feature 3: Risk ranking & officer review

Feature	What it means	Status
Risk score and colour	Warnings add up to GREEN, AMBER or RED	Done
Review queue	All colleges, highest risk first	Done
Review page	Summary, warnings with proof, verification, rules, map	Done
Accept or dismiss	The officer decides each warning and must write a reason; there is no reject button	Done
Decision support only	A banner on every officer and inspector page says so	Done

Smaller features that keep them working

Feature	What it means	Status
Document sorter	Says what each file is ("fire certificate, 97% sure") and finds each field	Done
Last year's records	A document already verified last year is not checked again	Done
Course & intake settings	The college sets its courses and seats; the norms are worked out from them	Done
Evidence viewer	Opens the document page with the exact spot highlighted	Done
Plain-language reasons	Every warning has a message in simple words	Done
Rule versions with dry-run	Test a rule change on every application before turning it on	Done
Rule-draft helper	An AI drafts a rule from Handbook text; a human must approve it (off unless switched on)	Done
Audit log	Every decision is logged in a chain that shows if anyone altered it	Done
Fix-list before submit	The college sees what to fix, in plain words, before submitting	Done
Alerts, PDF report, phone-app login	Missing info and expiry alerts, a full report with proof images, a code from a phone app for officers	Done

What each person gets

- **The inspector:** a ranked list instead of random picks, and a PDF on-site checklist. On site, the inspector marks each flagged item **Confirmed** or **Cleared** with a note, and a progress bar shows how much of the visit is done.
- **The college:** a fix-list before it submits. A college only ever sees its own application.

Why it is different

U1 Explainable	Every warning cites its Handbook clause and page
U2 Cross-checking	Facts are checked across documents, years and colleges
U3 Human in control	Decision support only, never auto approve or reject
U4 Audit	Tamper-evident, hash-chained log of every decision
U5 Prevention	Colleges fix problems before they submit
U6 Targeting	Red, amber and green tiers send inspectors to the riskiest colleges first

U7 Maintainability	Rules update without redeploying: an AI drafts, a human approves, a dry-run shows the effect

What exists today

Everything above is built and covered by the automated tests (backend suite plus the frontend build and smoke test), working against three made-up colleges: A is clean, B has two problems, C has many hidden ones (a land record edited after signing, 12 duplicate teachers, a copied photo). Optional extras stay switched off by default (scanned-page OCR, PC counting in lab photos, the AI rule-draft helper). All sources are fake and labelled MOCK.